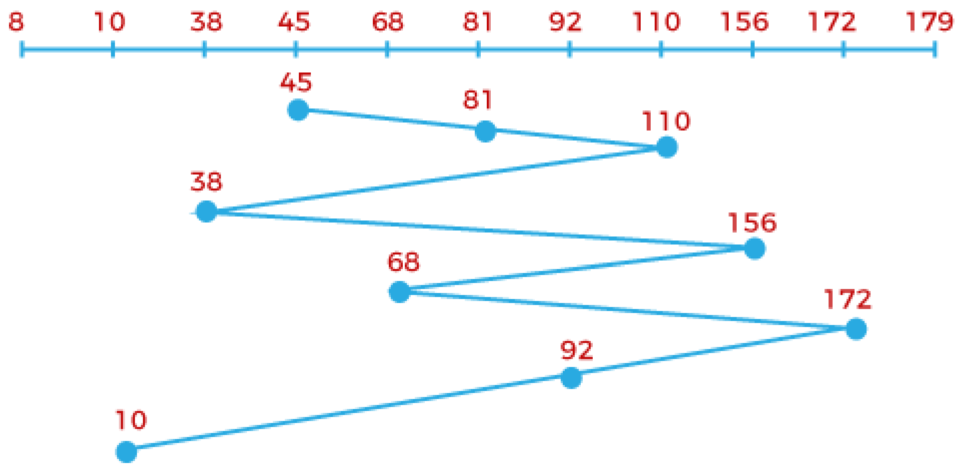
# HIRA KUMAR

**Q1. a disk with 180 tracks. order: 81, 110, 38, 156, 68, 172, 92, 10. R/w Head 45. Find FCFS.**

**Solution:→**




FCFS Disk Scheduling Algorithm

Total head movements,

The initial head point is **45**,

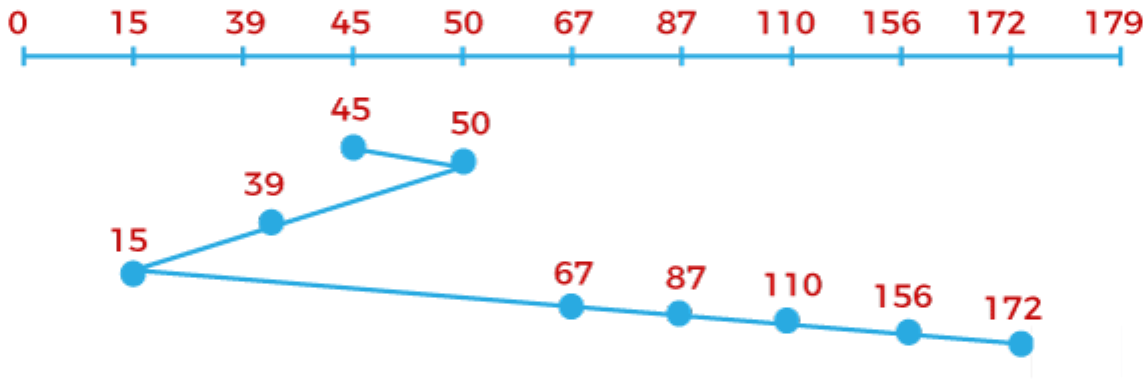
= (81-45) + (110-81) + (110-38) + (156-38)+ (156-68) + (172-68) + (172-92) + (92-10)

= 36 + 29 + 72 + 118 + 88 + 104 + 80 + 82

= 609

**Q2. a disk with 180 tracks order: 87, 110, 50, 172, 67, 156, 39, 15. Read/Write head is 45. Find SSTF**

**Solution:→**




SSTF Disk Scheduling Algorithm

Total head movements,

Initial head point is **45**,

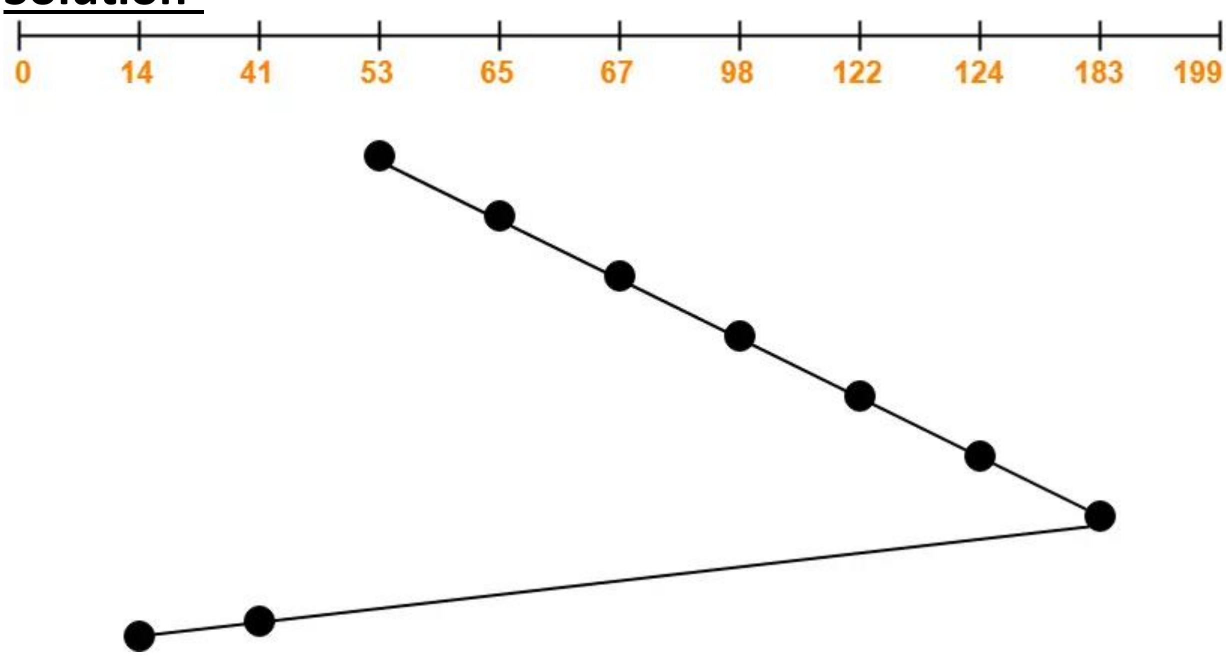
= (50-45) + (50-39) + (39-15) + (67-15) + (87-67) + (110-87) + (156-110) + (172-156)

= 5 + 11 + 24 + 52 + 20 + 23 + 46 + 16

= 197

**Q3. The cylinders are numbered from 0 to 199. Order 98,183, 41, 122, 14, 124, 65, 67. R/w head 53 moving towards larger cylinder numbers. Find LOOK algo.**

**Solution-**



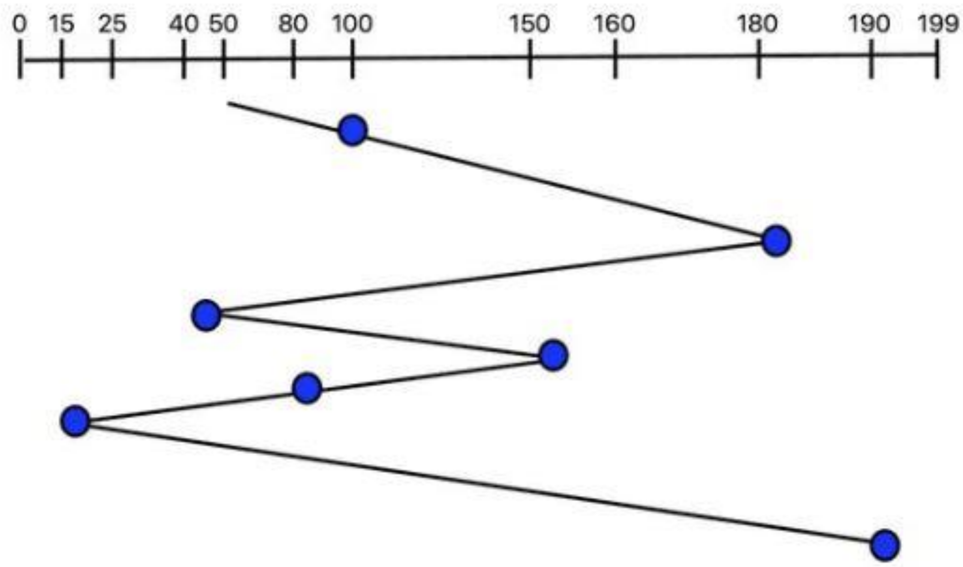

Total head movements incurred while servicing these requests

= (65 – 53) + (67 – 65) + (98 – 67) + (122 – 98) + (124 – 122) + (183 – 124) + (183 – 41) + (41 – 14)

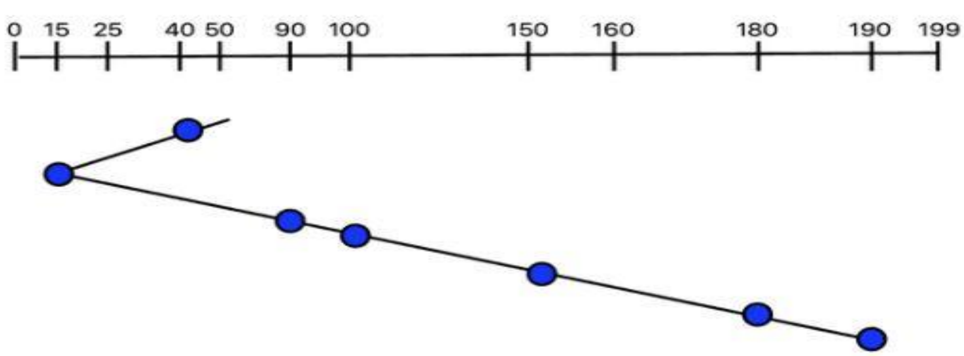
= 12 + 2 + 31 + 24 + 2 + 59 + 142 + 27
= 299

**Q4. Order → 100,180,40,150,80,15,190, r/w head – 50, Find FCFS & SSTF**

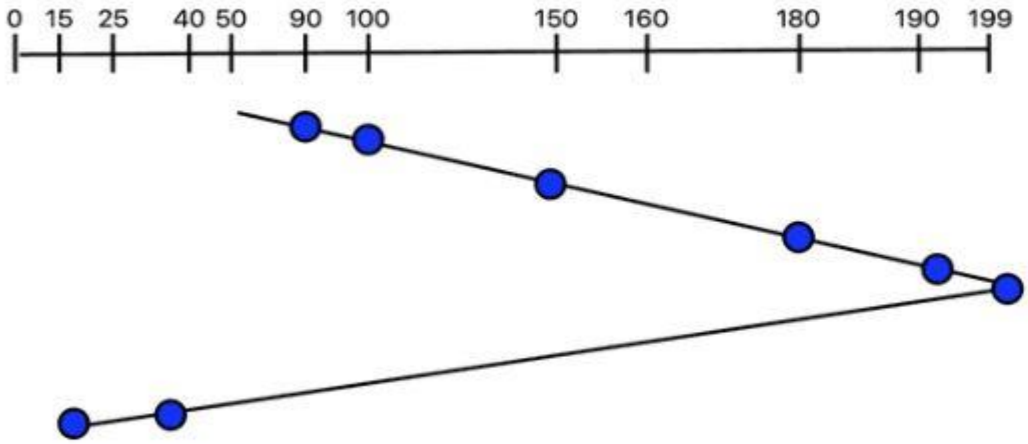
Hence, the total head movements for the FCFS algorithm above is given by = (100-50)+(180-100)+(180-40)+(150-40)+(150-80)+(80-15)+(190-15) = 690.



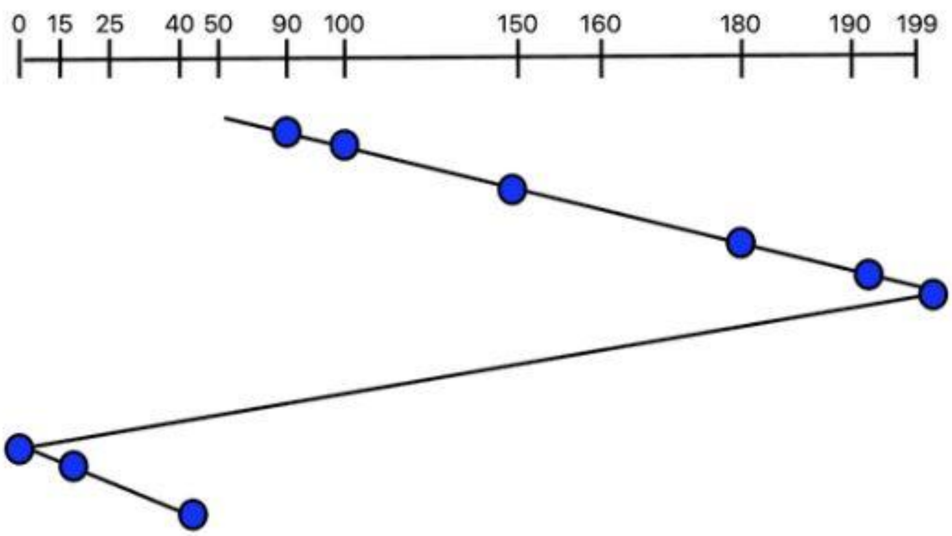

**SSTF =** The total head movement is given by = (50-40)+(40-15)+(90-15)+(100-90)+(150-100)+(180-150)+(190-180)=210.




**Q5 Order → 100,180,40,150,80,15,190, r/w head – 50 higher value, Find SCAN & CSCAN**

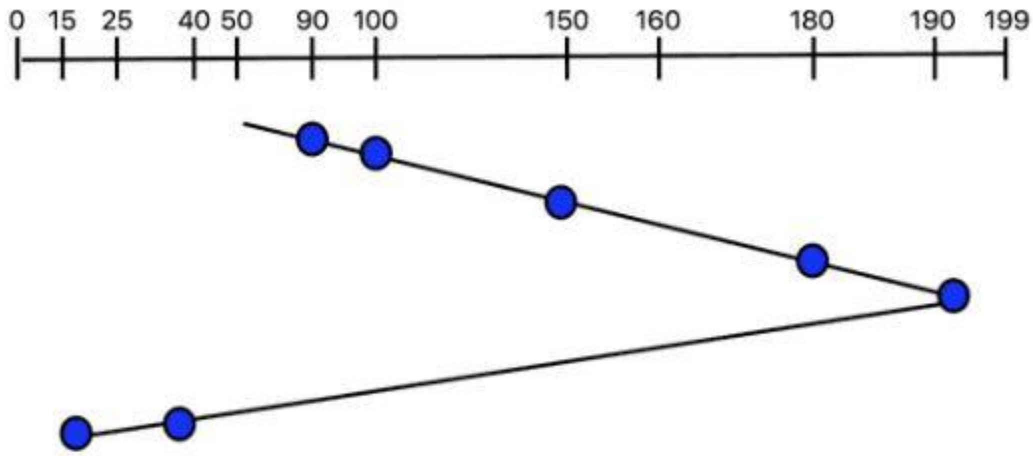



The total overhead movement is given by: (90-50)+(100-90)+(150-100)+(180-150)+(190-180)+(199-190)+(199-40)+(40-15)=333.
OR
We have: (199-50)+(199-15)=333

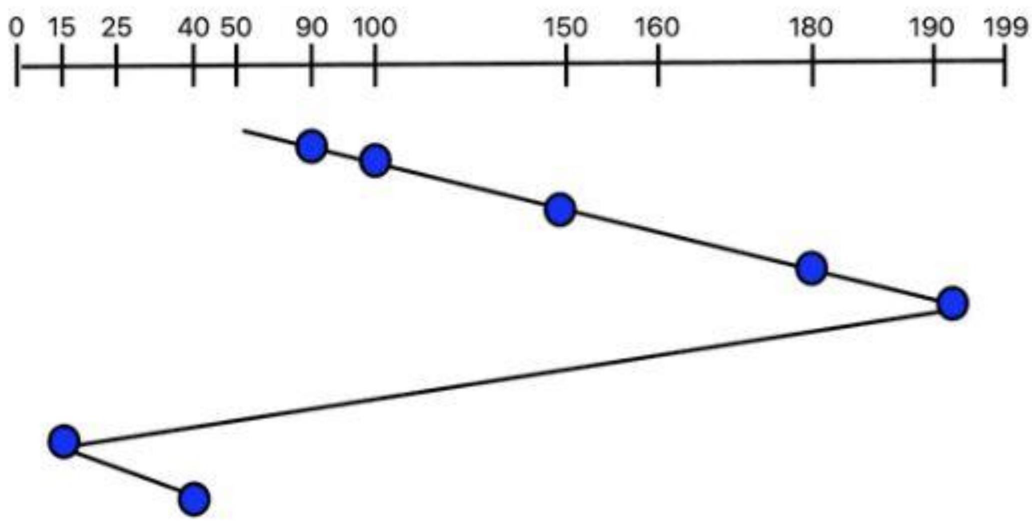



The total overhead movement is given by:(90-50)+(100-90)+(150-100)+(180-150)+(190-180)+(199-190)+(199-0)+(15-0)+(40-15)= 388.
Or simply: (199-50)+(199-0)+(40-0)=388

**Q6. Order → 100,180,40,150,80,15,190, r/w head – 50 higher value, Find LOOK & CLOOK**



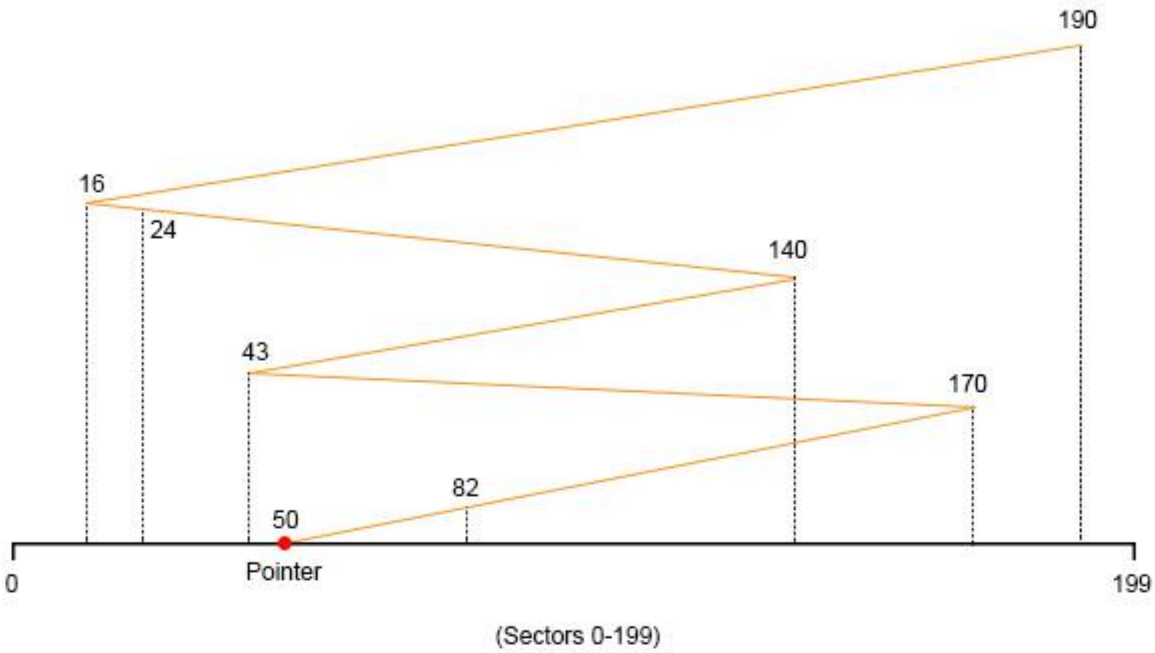

**LOOK** - The total overhead movement is simply given by: (190-50)+(190-15)=315




**CLOOK :-** The total overhead movement is given by: (190-50)+(190-15)+(40-15)=340

Q7. Request queue sequence is 82, 170, 43, 140, 24, 16, 190 respectively, for a disk with 200 tracks. R/W Head pointer is 50. Find total move in FCFS, SSTF & Find SCAN, CSCAN, LOOK, CLOOK for large value.

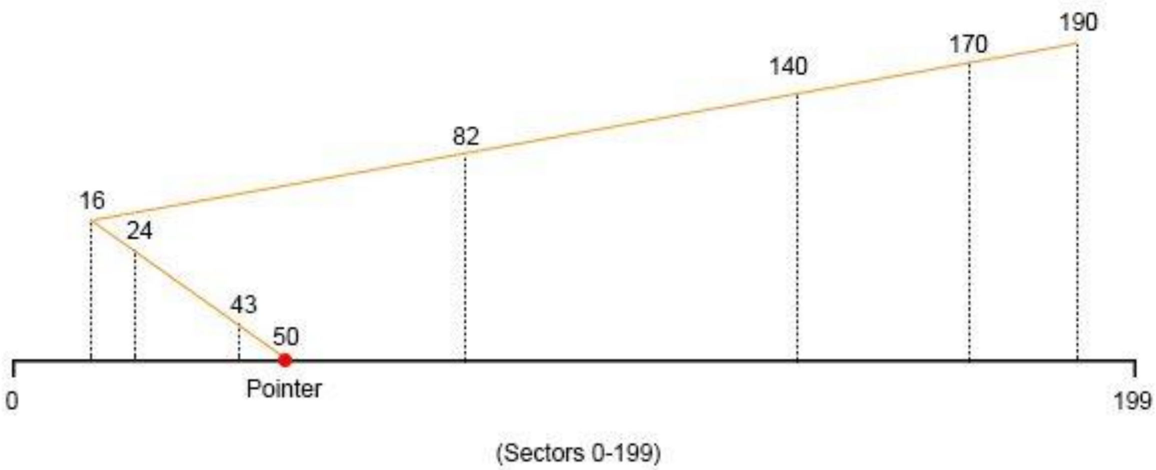
solution




Total Number of tracks/cylinders moved by the R/W head is
= (82-50)+(170-82)+(170-43)+(140-43)+(140-24)+(24-16)+(190-16)
= 32 + 88 + 127 + 97 + 116 + 8 + 174
= 642

# SSTF

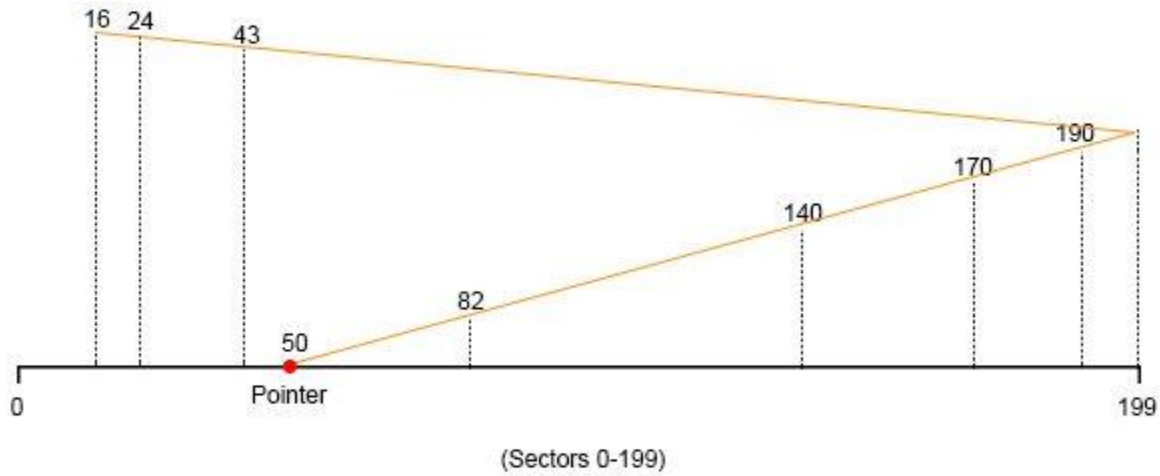
Solution




Total Number of tracks/cylinders moved by the R/W head is
= (50-43)+(43-24)+(24-16)+(82-16)+(140-82)+(170-140)+(190-170)
= 7 + 9 + 8 + 66 + 58 + 30 + 20
= 208

# SCAN

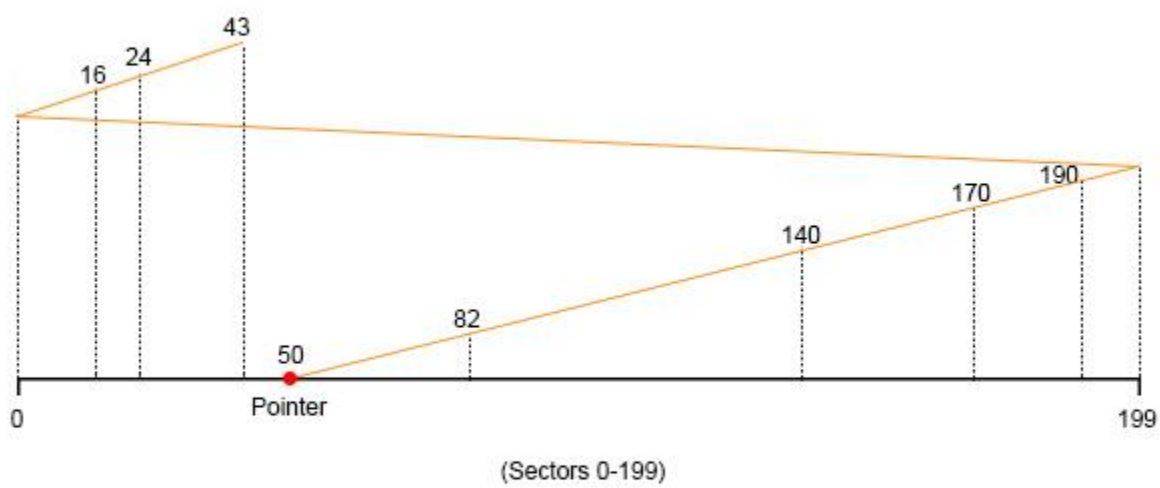
Solution




Total Number of tracks/cylinders moved by the R/W head is
= (82-50)+(140-82)+(170-140)+(190-170)+(199-190)+(199-43)+(43-24)+(24-16)
= 32 + 58 + 30 + 20 + 9 + 156 +19 +8
= 332

# CSCAN

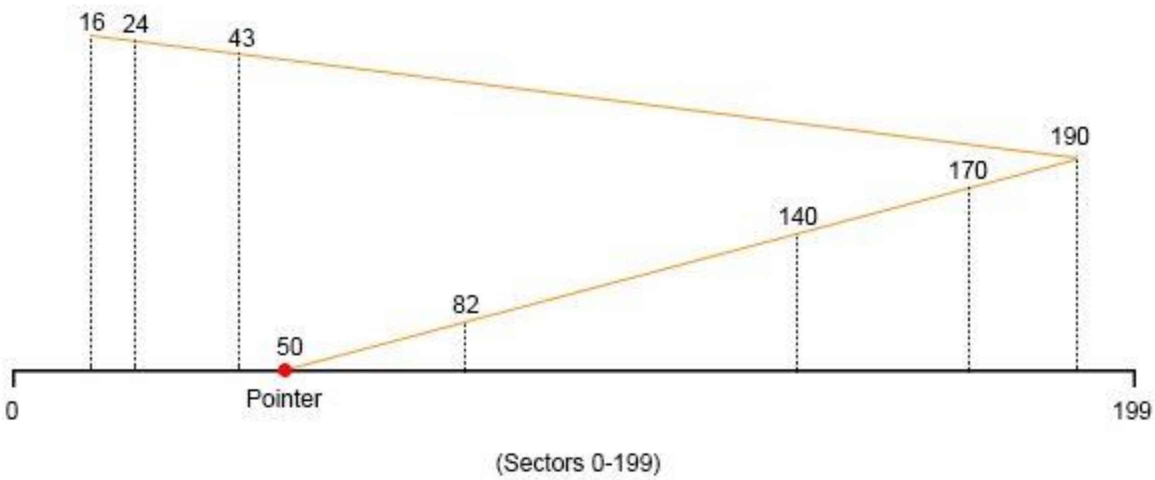
Solution




Total Number of tracks/cylinders moved by the R/W head is
= (82-50)+(140-82)+(170-140)+(190-170)+(199-190) + (199-0) +(16-0) +(24-16) +(43-24)
= 32 + 58 + 30 + 20 + 9 + 199 +16 +8 + 19
= 391

# LOOK

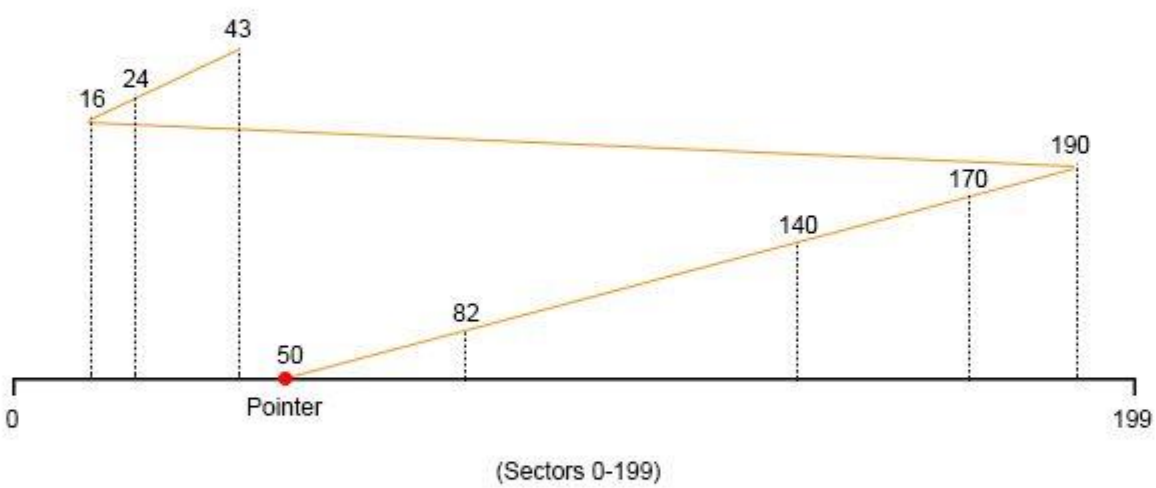
Solution




Total Number of tracks/cylinders moved by the R/W head is
= (82-50)+(140-82)+(170-140)+(190-170) +(190-43)+(43-24)+(24-16)
= 32 + 58 + 30 + 20 + 147 +19 +8
= 314

# CLOOK

Solution




Total Number of tracks/cylinders moved by the R/W head is
= (82-50) + (140-82) + (170-140) + (190-170) + (190-16) + (24-16) + (43-24)

$= 32 + 58 + 30 + 20 + 174 + 8 + 19$

$= 341$